Team racing rotations and boat pairings

The basic "final-four" boat rotation and pairings

In team racing championship rotations the boats carry an identifying characteristic (usually a shroud flag or pennant). One designated group of boats always races against a second designated group (for example, boats displaying white pennants always race against the boats displaying green pennants, red against blue, yellow against black, etc.). The teams in the boats usually change after every race. Ideally, each team sails each boat an equal number of times, although is not always possible unless a double round-robin is sailed. A shorthand method of referring to the rotations is by number of competing teams and number of boats used. For example, a 5-team, 6-boat rotation is referred to as "5x6"; an 11-team, 24-boat rotation is referred to as "11x24".

The system assumes that the boats are equal, so equalization is very important. "Fairness" is then furthered by designing the boat rotations and pairings in advance, and having the teams draw lots for their positions in the rotation. The championship rotations were originally written for the US SAILING Team Racing Championship (Hinman Trophy) by Mrs. Rich Treacy in 1987 for the 5x6 through 8x6, and 5x12 through 11x12. They have since been expanded for 18 and 24 boat rotations.

The basic team racing direct swap is shown in Table 5.21, in which four teams meet in a "double round-robin." The groups of boats (3 per group) are identified by four colored pennants—red, blue, green and white, although they can be any colors.

TABLE 5.21

Team Race Round Robin
4 Teams, 12 Boats, 4 Colors
12 Races(Double Round)

Race	Team	Pennant	Race	Team	Pennant
1	1	white	7	4	white
	4	green		1	green
2	2	red	8	3	red
	3	blue		2	blue
3	1	white	9	3	white
	3	green		1	green
4	2	red	10	4	red
	4	blue		2	blue
5	1	white	11	2	white
	2	green		1	green
6	3	red	12	4	red
	4	blue		3	blue

Running team races with limited numbers of boats

You can, however, run a team racing series with fewer than enough boats for all members of each team. In other words, 12 boats (4 groups, 3 boats each) will accommodate 4 teams. But if there are not enough boats, some of the teams may race while others stand by. The teams then exchange boats with the next teams to race.

Tables 5.22,5.23, 5.25,5.27 and 5.29 show rotations for round robins of races for 6 or 12 boats, for 5 through 9 teams, respectively.

A "limited rotation" variation on the "follow the leader" method

Where there are not enough boats to accommodate all of the competing teams, a "limited rotation" can also be used. For example, if ten teams are competing, but there are only enough boats for 6 teams, (18 boats), Ken Legler of Tufts University suggests the following round-robin format:

Six teams (A through F) sail the first three races. Upon finishing, teams A and B sail in to exchange with teams G and H, while teams C through F become the first two starts of the second round. Teams G and H become the third start of round 2. This pattern continues with teams staying on the water for either four or two straight races against other teams without exchanging boats, until it is their turn to go ashore and rotate out. Using this method, it is possible to complete all 45 races of a ten-team round-robin (see formula on page 5.27) in one day.

Pairings and rotations for round-robins

You may use the following Tables in lieu of designing a round-robin rotation. Each may be used by itself or in combination with other formats, such as a "best-of-three" final, a "final-four" round-robin, etc. (see "Suggested Formats" Section 3.4 above). As mentioned above, do not start with a rotation which is too ambitious. Seventy-eight races in a two day event is a realistic goal if your personnel expediting boat exchanges are experienced, and the race course is not far from the boat exchange location.
The Round-Robin Tables which follow are:

TABLE NO.	NO. OF TEAMS	NO. OF BOATS
5.22	5	12 (or 6)
5.23	6	12 (or 6)
5.24	7	12
5.25	7	12 (or 6)

TABLE NO.	NO. OF TEAMS	NO. OF BOATS
5.26	7	18
5.27	8	12 (or 6)
5.28	8	18
5.29	8	24
5.30	9	12
5.31	9	18
5.32	9	24
5.33	10	18
5.34	10	24
5.35	11	18
5.36	11	24
5.37	12	18
5.38	12	24

In each of the Tables, only a single round-robin is shown. A double round-robin is a repeat of the first round. For double round-robins, you may want to reverse the boat assignments so that at their next meeting in round 2, the competitors are sailing the boats their opponents sailed in round 1.

The comments after Tables 5.22,5.23,5.25 and 5.27 include suggestions how to change the Table if you are using only 6 boats. Although they are adaptable to either four groups (12 boats) or two (6 boats), eight teams is effectively the practical limit for using 6 boats, because the teams must wait so long between races.

TABLE 5.22

Team Race Round Robin 5 Teams, 12 (or 6) Boats, 4 Colors 10 Races(Single Round)					
Race	**Team**	**Pennant**	**Race**	**Team**	**Pennant**
1	1	white	6	1	red
	5	green		3	blue
2	1	red	7	2	white
	4	blue		5	green
3	2	white	8	3	red
	3	green		4	blue
4	1	red	9	1	white
	5	blue		2	green
5	2	white	10	4	red
	4	green		5	blue

Comments to Table 5.22:

1. For a double round-robin, repeat the same format for the second round. Time permitting, add a "best-of-three" final (see "Suggested Formats").
2. Where you have only 6 boats, the pairing is the same, but only two colors are used (for example, white and green). Instead of starting race 2 immediately after race 1, the teams must exchange boats after each race with the next teams to start.

TABLE 5.23

Team Race Round Robin
6 Teams, 12 (or 6) Boats, 4 Colors
15 Races(Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	1	white	9	5	white
	2	green		1	green
2	5	red	10	4	red
	3	blue		2	blue
3	4	white	11	6	white
	6	green		3	green
4	3	red	12	4	red
	1	blue		5	blue
5	2	white	13	2	white
	5	green		3	green
6	6	red	14	1	red
	1	blue		4	blue
7	3	white	15	5	white
	4	green		6	green
8	2	red			
	6	blue			

Comments to Table 5.23:

1. For a double round-robin, repeat the same format for the second round. Break between races 15 and 16, since the boats with the white and green pennants will be used in both races.
2. Instead, you may wish to use a single round-robin, followed by a "best-of-three" final, with all races counted (see "Suggested Formats").
3. Where you have only 6 boats, the pairing is the same, but only two colors are used (for example, white and green). Instead of starting race 2 immediately after race 1, the teams must exchange boats after each race with the next teams to start.
4. The boat rotation for six teams using 18 boats is only slightly quicker than the 12-boat rotation, so only the recommended 12-boat rotation is included here, between races 21 and 22, since the boats with the white and green pennants will be used in both races.
3. Where you have only 6 boats, the pairing is the same, but only two colors are used (for example, white and green). Instead of starting race 2 immediately after race 1, the teams must exchange boats after each race with the next teams to start.
4. Where you have 18 boats, use Table 5.26.

FIGURE 5.24

Team Racing Round-Robin
7 Teams, 12 Boats, 21 Races

Race Rotation:

Race:	Pairings:		
1	A (blue)*	v.	B (red)
2	C (white)	v.	D (green)
3	A (blue)	v.	C (white)
4	B (red)	v.	D (green)
5	A (white)	v.	E (blue)
6	F (red)	v.	G (green)
7	A (white)	v.	F (red)
8	E (blue) v.		G (green)
9	D (red)	v.	F (white)
10	B (blue)	v.	E (green)
11	B (blue)	v.	F (white)
12	D (red)	v.	E (green)
13	C (blue)	v.	F (green)
14	A (red)	v.	G (white)
15	C (blue)	v.	G (white)
16	A (red)	v.	D (green)
17	D (white)	v.	G (blue)
18	B (green)	v.	C (red)
19	B (green)	v.	G (blue)
20	C (red)	v.	E (white)
21	E (white)	v.	F (blue)

Sail-off race if necessary

	A	B	C	D	E	F	G	Total
A								
B								
C								
D								
E								
F								
G								

Final 8tandings:

1.

2.

3.

4.

5.

6.

Winning Team Personnel:

*** Indicates color of boat, pennant, etc.**

TABLE 5.25

Team Race Round Robin 7 Teams, 12 (or 6) Boats, 4 Colors 21 Races(Single Round)					
Race	**Team**	**Pennant**	**Race**	**Team**	**Pennant**
1	1	white	12	4	red
	3	green		7	blue
2	2	red	13	6	white
	4	blue		1	green
3	5	white	14	1	red
	7	green		2	blue
4	1	red	15	4	white
	6	blue		5	green
5	3	white	16	6	red
	4	green		7	blue
6	2	red	17	2	white
	7	blue		3	green
7	5	white	18	7	red
	6	green		1	blue
8	1	red	19	3	white
	4	blue		5	green
9	7	white	20	4	red
	3	green		6	blue
10	6	red	21	5	white
	2	blue		2	green
11	5	white			
	1	green			

Comments to Table 5.25:

1. Table 5.25 is the same type of format as Table 5.24, but expressed differently.

2. You may wish to use a single round-robin, followed by a "final four" round robin. Table 5.21 (see "Suggested Formats").

3. For a double round-robin, repeat the same format for the second round. Break between races 21 and 22, since the boats with the white and green pennants will be used in both races.

3. Where you have only 6 boats, the pairing is the same, but only two colors are used (for example, white and green). Instead of starting race 2 immediately after race 1, the teams must exchange boats after each race with the next teams to start.

5. Where you have 18 boats, use Table 5.26.

TABLE 5.26

Team Race Round Robin
7 Teams, 18 Boats, 6 Colors
21 Race (SingleRound)

Race	Team	Pennant	Race	Team	Pennant
1	1	white	12	4	yellow
	3	green		7	black
2	4	red	13	3	white
	2	blue		6	green
3	5	yellow	14	2	red
	7	black		1	blue
4	6	white	15	5	yellow
	1	green		4	black
5	3	red	16	7	white
	4	blue		6	green
6	7	yellow	17	2	white
	2	black		3	green
7	5	white	18	2	red
	6	green		3	blue
8	1	red	19	5	white
	4	blue		3	green
9	3	yellow	20	4	red
	7	black		6	blue
10	6	white	21	2	yellow
	2	green		5	black
11	5	red			
	1	blue			

TABLE 5.27

Team Race Round Robin
8 Teams, 12 (or 6) Boats, 4 Colors
28 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	1	white	15	3	white
	3	green		4	green
2	2	red	16	7	red
	6	blue		8	blue
3	7	white	17	5	white
	4	green		6	green
4	5	red	18	1	red
	8	blue		4	blue
5	6	white	19	2	white
	1	green		7	green
6	3	red	20	3	red
	7	blue		8	blue
7	2	white	21	4	white
	4	green		5	green
8	1	red	22	6	red
	5	blue		7	blue
9	6	white	23	1	white
	8	green		2	green
10	2	red	24	5	red
	3	blue		3	blue
11	5	white	25	4	white
	7	green		8	green
12	4	red	26	7	red
	6	blue		1	blue
13	8	white	27	3	white
	1	green		6	green
14	2	red	28	8	red
	5	blue		2	blue

Comments Table 5.27

1. For a double round-robin, repeat the same format for the second round.
2. Where teams are unevenly matched, use a bracket system elimination (Table 5.16). Where they are evenly matched, use either a double round-robin, or a single round-robin followed by either a "best-of-three" final or a "final-four" round-robin (Table 5.21)
3. Eight teams is effectively the practical limit for using only 6 boats, because of the amountof time each team must wait to race. Where you have only 6 boats, the pairing is the same, but only two colors are used (for example, white and green), Instead of starting race two immediately after race 1, the teams must exchange boats after each race with the next teams to start.

TABLE 5.28

Team Race Round Robin 8 Teams, 18 Boats, 6 Colors 28 Races (Single Round)					
Race	**Team**	**Pennant**	**Race**	**Team**	**Pennant**
1	1	white	15	3	yellow
	2	green		8	black
2	3	red	16	6	white
	4	blue		1	green
3	6	yellow	17	4	red
	5	black		7	blue
4	7	white	18	8	yellow
	8	green		2	black
5	1	red	19	3	white
	3	blue		5	green
6	2	yellow	20	7	red
	4	black		1	blue
7	7	white	21	4	yellow
	5	green		8	black
8	6	red	22	2	white
	8	blue		6	green
9	1	yellow	23	3	red
	4	black		7	blue
10	5	white	24	5	yellow
	2	green		1	black
11	3	red	25	4	white
	6	blue		6	green
12	8	yellow	26	2	red
	1	black		3	blue
13	2	white	27	5	yellow
	7	green		8	black
14	4	red	28	6	white
	5	blue		7	green

TABLE 5.29

Team Race Round Robin
8 Teams, 24 Boats, 8 Colors
28 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	1	white	15	3	yellow
	2	green		7	black
2	3	red	16	4	orange
	4	blue		8	light blue
3	5	yellow	17	1	white
	6	black		6	green
4	7	orange	18	3	red
	8	light blue		8	blue
5	1	white	19	2	yellow
	3	green		5	black
6	2	red	20	4	orange
	4	blue		7	light blue
7	5	yellow	21	1	white
	7	black		7	green
8	6	orange	22	2	red
	8	light blue		8	blue
9	1	white	23	3	yellow
	4	green		5	black
10	2	red	24	4	orange
	3	blue		6	light blue
11	5	yellow	25	1	white
	8	black		8	green
12	6	orange	26	2	red
	7	light blue		7	blue
13	1	white	27	3	yellow
	5	green		6	black
14	2	red	28	4	orange
	6	blue		5	light blue

Comment to Tables 5.33-5.38:
Use these single round robins with either a "best-of-three" final or a "final-four" round robin (see "Suggested Formats", "Basic final-Four" and Table 5.21 above).

TABLE 5.30

Team Race Round Robin
9 Teams, 12 Boats, 4 Colors
36 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	5	white	19	9	white
	6	green		6	green
2	3	red	20	5	red
	4	blue		8	blue
3	7	white	21	2	white
	8	green		4	green
4	9	red	22	1	red
	1	blue			blue
5	2	white	23	7	white
	6	green		9	green
6	3	red	24	6	red
	5	blue		8	blue
7	8	white	25	1	white
	1	green		5	green
8	9	red	26	2	red
	4	blue		3	blue
9	2	white	27	4	white
	5	green		7	green
10	7	red	28	8	red
	6	blue		9	blue
11	3	white	29	6	white
	9	green		3	green
12	4	red	30	2	red
	8	blue		1	blue
13	6	blue	31	5	white
	1	green		9	green
14	5	red	32	4	red
	7	blue		6	blue
15	9	white	33	3	white
	2	green		7	green
16	1	red	34	8	red
	4	blue		2	blue
17	8	white	35	4	white
	3	green		5	green
18	7	red	36	1	red
	2	blue		7	blue

TABLE 5.31

Team Race Round Robin
9 Teams, 18 Boats, 6 Colors
36 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	6	white	19	9	white
	5	green		6	green
2	3	red	20	5	red
	4	blue		8	blue
3	8	yellow	21	2	yellow
	7	black		4	black
4	9	white	22	1	white
	1	green		3	green
5	6	red	23	7	red
	2	blue		9	blue
6	3	yellow	24	6	yellow
	5	black		8	black
7	8	white	25	1	white
	1	green		8	green
8	9	red	26	2	red
	4	blue		3	blue
9	5	yellow	27	4	yellow
	2	black		7	black
10	7	white	28	8	white
	6	green		9	green
11	3	red	29	6	red
	9	blue		3	blue
12	4	yellow	30	2	yellow
	8	black		1	black
13	6	white	31	5	white
	1	green		9	green
14	5	red	32	4	red
	7	blue		6	blue
15	9	yellow	33	7	yellow
	2	black		3	black
16	1	white	34	2	white
	4	green		8	green
17	8	red	35	4	red
	3	blue		5	blue
18	7	yellow	36	1	yellow
	2	black		7	black

TABLE 5.32

Team Race Round Robin
9 Teams, 24 Boats, 8 Colors
28 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	1	white	19	2	yellow
	2	green		9	black
2	3	red	20	4	orange
	4	blue		7	light blue
3	5	yellow	21	1	white
	6	black		7	green
4	7	orange	22	2	red
	8	light blue		8	blue
5	1	white	23	9	yellow
	3	green		5	black
6	2	red	24	4	orange
	4	blue		6	light blue
7	5	yellow	25	1	white
	7	black		8	green
8	6	orange	26	2	red
	9	light blue		7	blue
9	1	white	27	3	yellow
	4	green		6	black
10	2	red	28	4	orange
	3	blue		9	light blue
11	5	yellow	29	1	white
	8	black		9	green
12	7	orange	30	6	red
	9	light blue		8	blue
13	1	white	31	3	yellow
	5	green		7	black
14	2	red	32	2	orange
	6	blue		5	light blue
15	3	yellow	33	3	white
	9	black		5	green
16	4	orange	34	6	red
	8	light blue		7	blue
17	1	white	35	8	yellow
	6	green		9	black
18	3	red	36	4	orange
	8	blue		5	light blue

Taken from "***Race Management Handbook***", Chapter 5 © 2001, US SAILING

TABLE 5.33

Team Race Round Robin
10 Teams, 18 Boats, 6 Colors
45 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant	Race	Team	Pennant	Race	Team	Pennant
1	1 6	white green	13	3 10	white green	25	9 5	white green	37	10 8	white green
2	7 2	red blue	14	4 5	red blue	26	6 2	red blue	38	4 7	red blue
3	3 8	yellow black	15	7 6	yellow black	27	1 3	yellow black	39	5 1	yellow black
4	9 4	white green	16	2 1	white green	28	8 7	white green	40	6 9	white green
5	5 10	red blue	17	9 8	red blue	29	10 9	red blue	41	2 10	red blue
6	1 7	yellow black	18	3 4	yellow black	30	4 2	yellow black	42	1 4	yellow black
7	8 2	white green	19	5 7	white green	31	3 5	white green	43	7 3	white green
8	3 9	red blue	20	10 6	red blue	32	8 6	red blue	44	5 8	red blue
9	10 4	yellow black	21	9 1	yellow black	33	10 1	yellow black	45	4 6	yellow black
10	5 6	white green	22	2 3	white green	34	7 9	white green			
11	1 8	red blue	23	8 4	red blue	35	2 5	red blue			
12	2 9	yellow black	24	7 10	yellow black	36	6 3	yellow black			

Taken from "***Race Management Handbook***", Chapter 5 © 2001, US SAILING

TABLE 5.34

Team Race Round Robin
10 Teams, 18 Boats, 6 Colors
45 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant	Race	Team	Pennant	Race	Team	Pennant
1	1 6	white green	13	3 10	white green	25	9 5	white green	37	10 8	white green
2	7 2	red blue	14	4 5	red blue	26	6 2	red blue	38	4 7	red blue
3	3 8	yellow black	15	7 6	yellow black	27	1 3	yellow black	39	5 1	yellow black
4	9 4	orange light blue	16	2 1	orange light blue	28	8 7	orange light blue	40	6 9	orange light blue
5	5 10	white green	17	9 8	white green	29	10 9	white green	41	2 10	white green
6	1 7	red blue	18	3 4	red blue	30	4 2	red blue	42	1 4	red blue
7	8 2	yellow black	19	5 7	yellow black	31	3 5	yellow black	43	7 3	yellow black
8	3 9	orange light blue	20	10 6	orange light blue	32	8 6	orange light blue	44	5 8	orange light blue
9	10 4	white green	21	9 1	white green	33	10 1	white green	45	4	white green
10	5 6	red blue	22	2 3	red blue	34	7 9	red blue			
11	1 8	yellow black	23	8 4	yellow black	35	2 5	yellow black			
12	2 9	orange light blue	24	7 10	orange light blue	36	6 3	orange light blue			

TABLE 5.35

Team Race Round Robin
11 Teams, 18 Boats, 6 Colors
55 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	6	white	15	5	yellow
	1	green		9	black
2	2	red	16	11	white
	7	blue		1	green
3	8	yellow	17	2	red
	3	black		10	blue
4	4	white	18	6	yellow
	9	green		3	black
5	10	red	19	4	white
	5	blue		7	green
6	11	yellow	20	5	red
	6	black		8	blue
7	7	white	21	9	yellow
	1	green		1	black
8	8	red	22	3	white
	2	light blue		11	green
9	3	yellow	23	6	red
	9	black		10	black
10	4	white	24	5	yellow
	10	green		2	black
11	5	red	25	1	white
	11	blue		4	green
12	2	yellow	26	7	red
	6	black		8	blue
13	3	white	27	9	yellow
	7	green		10	black
14	4	red	28	2	white
	8	blue		3	green

Table 3.35 Continued

Team Race Round Robin
11 Teams, 18 Boats, 6 Colors
55 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
29	6	red	43	9	white
	5	blue		7	green
30	11	yellow	44	8	red
	4	black		1	blue
31	1	white	45	10	yellow
	8	green		11	black
32	9	red	46	1	white
	2	blue		3	green
33	10	yellow	47	2	red
	3	black		3	blue
34	7	white	48	9	yellow
	6	green		8	black
35	4	red	49	7	white
	5	blue		11	green
36	8	yellow	50	10	red
	11	black		1	blue
37	1	white	51	3	yellow
	2	green		5	black
38	9	red	52	6	white
	6	blue		2	green
39	7	yellow	53	11	red
	10	black		9	blue
40	3	white	54	10	yellow
	4	green		8	black
41	11	red	55	5	white
	2	blue		7	green
42	1	yellow			
	5	black			

TABLE 5.36

Team Race Round Robin
11 Teams, 24 Boats, 8 Colors
55 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	6	white	15	5	yellow
	1	green		9	black
2	2	red	16	11	orange
	7	blue		1	light blue
3	8	yellow	17	2	white
	3	black		10	green
4	4	orange	18	6	red
	9	light blue		3	blue
5	10	white	19	4	yellow
	5	green		7	black
6	11	red	20	5	orange
	6	blue		7	light blue
7	7	yellow	21	9	white
	1	black		1	green
8	8	orange	22	3	red
	2	light blue		11	blue
9	3	white	23	6	yellow
	9	green		10	black
10	4	red	24	5	orange
	10	blue		2	light blue
11	5	yellow	25	1	white
	11	black		4	green
12	3	orange	26	7	red
	7	light blue		8	blue
13	1	white	27	9	yellow
	5	green		10	black
14	4	red	28	2	orange
	8	blue		3	light blue

continued, next page–

TABLE 5.36 continued

Team Race Round Robin
11 Teams, 24 Boats, 8 Colors
55 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
29	6	white	43	9	yellow
	51	green		7	black
30	11	red	44	8	orange
	4	blue		6	light blue
31	1	yellow	45	10	white
	8	black		11	green
32	9	orange	46	1	red
	2	light blue		3	blue
33	10	white	47	2	yellow
	3	green		4	black
34	7	red	48	9	orange
	6	blue		8	light blue
35	4	yellow	49	7	white
	5	black		11	green
36	8	orange	50	10	red
	11	light blue		1	blue
37	1	white	51	3	yellow
	2	green		5	black
38	9	red	52	5	orange
	6	blue		2	light blue
39	7	yellow	53	11	white
	10	black		9	green
40	3	orange	54	10	red
	4	light blue		8	blue
41	11	white	55	5	yellow
	2	green		7	black
42	1	red			
	5	blue			

TABLE 5.37

Team Race Round Robin
12 Teams, 18 Boats, 6 Colors
66 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	1	white	18	8	yellow
	2	green		11	black
2	3	red	19	5	white
	4	blue		1	green
3	5	yellow	20	2	red
	6	black		6	blue
4	7	white	21	3	yellow
	8	green		7	black
5	9	red	22	4	white
	10	blue		8	green
6	11	yellow	23	9	red
	12	black		8	blue
7	1	white	24	6	yellow
	3	green		10	black
8	4	red	25	7	white
	2	blue		11	green
9	5	yellow	26	8	red
	7	black		12	blue
10	6	white	27	2	yellow
	8	green		3	black
11	9	red	28	5	white
	11	blue		4	green
12	10	yellow	29	6	red
	12	black		7	blue
13	4	white	30	9	yellow
	1	green		8	black
14	5	red	31	11	white
	2	blue		10	green
15	3	yellow	32	1	red
	6	black		12	blue
16	10	white	33	3	yellow
	7	green		5	black
17	8	red	34	4	white
	11	blue		6	green

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TABLE 5 . 37 continued

Team Race Round Robin 12 Teams, 18 Boats, 6 Colors 66 Races (Single Round)					
Race	**Team**	**Pennant**	**Race**	**Team**	**Pennant**
35	7	red	51	12	yellow
	9	blue		2	black
36	10	yellow	52	11	white
	8	black		4	green
37	2	white	53	9	red
	11	green		3	blue
38	1	red	54	8	yellow
	6	blue		1	black
39	7	yellow	55	2	white
	12	black		10	green
40	4	white	56	6	red
	9	green		12	blue
41	3	red	57	11	yellow
	8	blue		5	black
42	5	yellow	58	7	white
	10	black		4	green
43	6	white	59	10	red
	11	green		1	blue
44	2	red	60	2	yellow
	7	blue		9	black
45	1	yellow	61	3	white
	9	black		11	green
46	12	white	62	12	red
	4	green		5	blue
47	5	red	63	4	yellow
	8	blue		10	black
48	10	yellow	64	8	white
	3	black		2	green
49	9	white	65	11	red
	6	green		1	blue
50	7	red	66	12	yellow
	1	blue		3	black

TABLE 5.38

Team Race Round Robin
12 Teams, 24 Boats, 8 Colors
66 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
1	1	white	18	8	red
	2	green		11	blue
2	3	red	19	5	yellow
	4	blue		1	black
3	5	yellow	20	2	orange
	6	black		6	light blue
4	7	orange	21	3	white
	8	light blue		7	green
5	9	white	22	4	red
	10	green		8	blue
6	11	red	23	9	yellow
	12	blue		5	black
7	1	yellow	24	6	orange
	3	black		10	light blue
8	4	orange	25	7	white
	2	light blue		11	green
9	5	white	26	8	red
	7	green		12	blue
10	6	red	27	2	yellow
	8	blue		3	black
11	9	yellow	28	5	orange
	11	black		4	light blue
12	10	orange	29	6	white
	12	light blue		7	green
13	4	white	30	9	red
	1	green		8	blue
14	5	red	31	11	yellow
	2	blue		10	black
15	3	yellow	32	1	orange
	6	black		12	light blue
16	10	orange	33	3	white
	7	light blue		5	green
17	8	white	34	4	red
	11	green		6	blue

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TABLE 5 . 38 continued

Team Race Round Robin
12Teams, 24 Boats, 8 Colors
66 Races (Single Round)

Race	Team	Pennant	Race	Team	Pennant
35	7	yellow	51	12	yellow
	9	black		2	black
36	10	orange	52	11	orange
	8	light blue		4	light blue
37	2	white	53	9	white
	11	green		3	green
38	2	red	54	8	red
	6	blue		1	blue
39	7	yellow	55	2	yellow
	12	black		10	black
40	4	orange	56	6	orange
	9	light blue		12	light blue
41	3	white	57	11	white
	8	green		5	green
42	5	red	58	7	red
	10	blue		4	blue
43	6	yellow	59	10	yellow
	11	black		1	black
44	2	orange	60	2	orange
	7	light blue		9	light blue
45	1	white	61	3	white
	9	green		11	green
46	12	red	62	12	red
	4	blue		5	blue
47	5	yellow	63	4	yellow
	8	black		19	black
48	10	orange	64	8	orange
	3	light blue		2	light blue
49	9	white	65	11	white
	6	green		1	green
50	4	red	66	12	red
	1	blue		3	blue

The easiest team racing events to run are bring-your-own-boat (BYOB) events. Boats do not have to be acquired, equalized, assigned, rotated, repaired or worried about. Also, the racing does not need to be run near a rotation dock and races do not need to be postponed for breakdowns. Even the rotation is easier to figure. In the following rotations there is a space in between groups of races or flights.

TABLE 5.39

3 teams	4 teams	5 teams	6 teams	7 teams
1. AB	1. AB	1. AB	1. AB	1. AB
	2. CD	2. CD	2. CD	2. CD
2. AC			3. EF	3. EF
	3. AC	3. AE		
3. BC	4. BD	4. CB	4. AC	4. AG
			5. BE	5. BC
	5. AD	5. AC	6. DF	6. DE
	6. BC	6. DE		
			7. AE	7. FG
		7. BE	8. CF	8. AC
		8. AD	9. BD	9. BD
		9. CE	10. AF	10. EG
		10. BD	11. DE	11. CF
			12. BC	12. AD
			13. AD	13. BG
			14. BF	14. CE
			15. CE	15. AF
				16. DG
				17. BF
				18. AE
				19. CG
				20. DF
				21. BE